SOLUTION
SECTION-I

Single Correct Answer Type **9 Q. [3 M (–1)]**

1. Ans. (B)

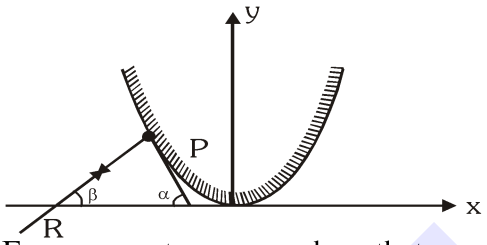
2. Ans. (B)

Sol. $\mu_1 \sin i = \mu_2 \sin 90°$

$$\tan i = \frac{\sqrt{\left(2^2+\left(\sqrt{5}\right)^2\right)}}{4} \Rightarrow i = 37° \Rightarrow \mu_2 = \frac{6}{5}$$

3. Ans. (A)

Sol. Point of intersection of two curves, $y = \frac{x^2}{4}$ and $y = x + 3$ is, $(-2, 1)$



$$\frac{dy}{dx} = \frac{x}{2} \text{ i.e., } \left(\frac{dy}{dx}\right)_{x=-2} = \frac{-2}{2} = -1$$

$\tan \beta$ = slope of incident ray RP $= 1$ or $\beta = 45°$

and $\tan\alpha = \left|\left(\frac{dy}{dx}\right)_{x=-2}\right| = 1 \Rightarrow \alpha = 45°$

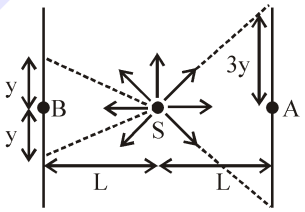
From geometry we can show that $\angle i = \angle r = 0°$ i.e., the ray is incident normally.

Hence, the desired unit vector : $A = \frac{1}{\sqrt{2}}\left(-\hat{i} - \hat{j}\right)$

4. Ans. (A)

Sol. $I_{net} = I_A + I_B$

$$I_B = \frac{I_0}{4\pi L^2 \times 9}, I_A = \frac{P_0}{4\pi L^2} = 8.1\mu$$



$$I_{net} = 8.1\ \mu + \frac{8.1\mu}{9}$$

$$I_{net} = 9\ \mu$$

5. Ans. (C)

Sol. For the image to coincide with object, image formed by concave mirror must be at 'a' in front of plane mirror.

$$\therefore \frac{1}{v}+\frac{1}{u}=\frac{1}{f} \Rightarrow \frac{-1}{2a}-\frac{1}{4a}=\frac{1}{f} \Rightarrow f = -4$$

$\therefore$ R = 8m

6. Ans. (A)

Sol. At t = 3 sec the radius becomes ∞ and the mirror behaves like plane mirror.

$$\frac{V_I + V_0}{2} = V_M$$

$$\frac{V_I + 6}{2} = -3$$

$$V_I = -12$$

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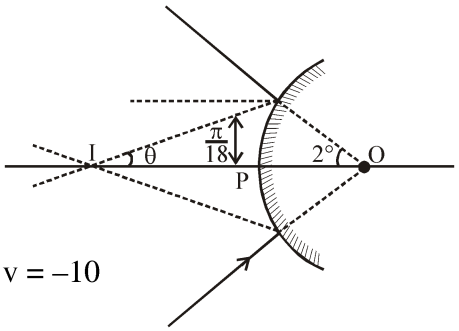
7. Ans. (D)

Sol. $\frac{\frac{\pi}{18}}{OP} = 2 \times \frac{\pi}{180}$

$OP = 5$

$\frac{1}{v} + \frac{1}{5} = \frac{1}{10}$

$f = 10$

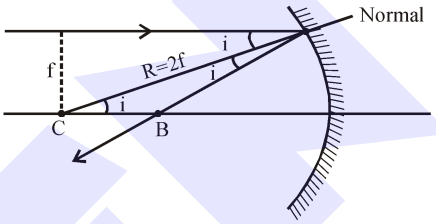


$\theta = \frac{\pi}{18 \times 10} = \frac{\pi}{180} = 1°$

8. Ans. (A)

Sol. $\sin i = \frac{f}{R} = \frac{1}{2}$; $i = 30°$

Not a paraxial ray



$CB = \frac{R}{2\cos i} = \frac{f}{(\sqrt{3}/2)} = f\left(\frac{2}{\sqrt{3}}\right)$

$\frac{CB}{f} = \frac{2}{\sqrt{3}}$

9. Ans. (D)

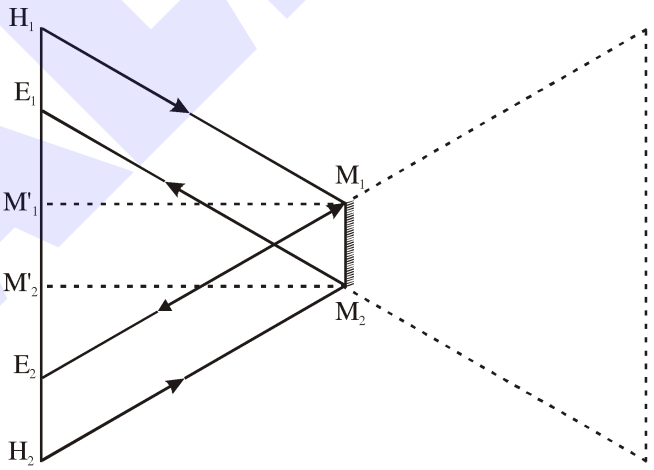
Sol. Let's say $H_1H_2 = D$, $E_1E_2 = d$

H_1H_2 = width of persons face.

E_1E_2 = separation between the eyes.

Let's say M_1M_2 = Width of mirror required.

Then following will be the ray diagram and calculations.



$H_1M'_1 = \frac{H_1E_2}{2}$ & $H_2M'_2 = \frac{H_2E_1}{2}$

$H_1E_2 = D - \frac{1}{2}(D-d) = \frac{D+d}{2} = H_2E_1$

$M'_1M'_2 = D - H_1M'_1 - H_2M'_2$

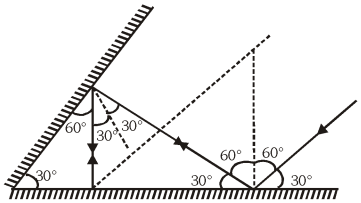
$= D - \left(\frac{D+d}{2}\right) = \frac{D-d}{2}$

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Multiple Correct Answer Type 8 Q. [4 M (–1)]

10. Ans. (A,C)

Sol.



11. Ans. (B,C,D)

Sol. y component of velocity is changing

Mirror must be in x-z plane

Since ray was initially travelling towards +y axis

Normal must be along –ve y i.e. $-\hat{j}$

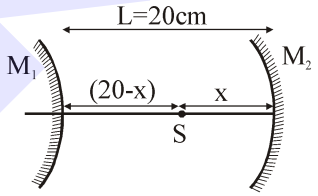
$\tan\theta = \frac{3}{4}$ angle of incidense = 37°

12. Ans. (A,C)

Sol. For first reflection from convex surface

$u_1 = -(20-x)$

$f_1 = +6$



$$v_1 = \frac{u_1 f_1}{u_1 - f_1} = \frac{(20-x)6}{26-x}$$

and for second reflection from concave surface,

$v_2 = (-x)$

$f_2 = -6$

$$\frac{1}{-x} + \frac{1}{u_2} = \frac{1}{-6} \Rightarrow \frac{1}{u_2} = \frac{1}{x} - \frac{1}{6}$$

$$u_2 = \frac{6x}{6-x}$$

But image of first reflection is object for second reflection,

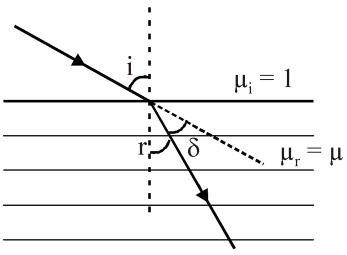
so, $u_2 = v_1 + 20$

or $\frac{6x}{6-x} = \left[\frac{(20-x)6}{26-x}\right] + 20 \Rightarrow x = 8$ cm

By principle of reversibility of light, it is also, true for first reflection on concave mirror and then on convex mirror.

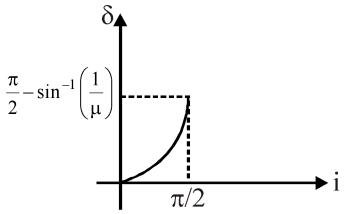
13. Ans. (A,B,D)

Sol. Light travels from rarer to denser medium



$\delta = i - r$

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$\delta = i - \sin^{-1}\dfrac{(\sin i)}{\mu}$ non-linear increasing function of i.

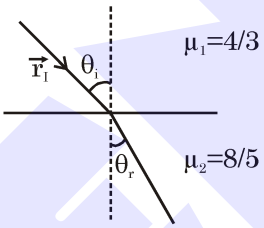
slope of tangent : $\dfrac{d\delta}{d_i} = 1 - \dfrac{\cos i}{\sqrt{\mu^2 - \sin^2 i}}$

Slope of tangent increases as i increases

Hence 'c' is not the graph for light that travels from rarer to denser medium.

14. Ans. (A,C)

Sol. $\vec{r}_I = +3\hat{i} - 4\hat{j}$



$\tan\theta_i = \dfrac{3}{4}$

$\theta_i = 37°$

$\mu_1 \sin\theta_i = \mu_2 \sin\theta_r$

$\dfrac{4}{3}\times\dfrac{3}{5} = \dfrac{8}{5}\sin\theta_r$

$\sin\theta_r = \dfrac{1}{2}$ $\theta_r = 30°$

15. Ans. (B,D)

Sol. Let the critical angle of interface between media 1 and 2 is C_1 and between 1 and 3 is C_2.

Then $\sin C_1 = \dfrac{n_2}{n_1}$ and $\sin C_2 = \dfrac{n_3}{n_1}$.

From TIR at second interface $90 - C_1 > C_2$. taking sin of both side we get

$\cos C_1 > \sin C_2$ or $\sqrt{1-\left(\dfrac{n_2}{n_1}\right)^2} > \dfrac{n_3}{n_1}$ or $n_1^2 - n_3^2 > n_2^2$

16. Ans. (B,C)

Sol. $m^2 = \left(\dfrac{v-f}{f}\right)^2$ = graph will be a parabola

m = 0 at focus $\Rightarrow$ f = 30

b =1 = m at v = 0 = a as mag at 2f

at v = 15 m = $\dfrac{1}{2}$; $m^2 = \dfrac{1}{4} \Rightarrow \dfrac{a}{4} = c$

17. Ans. (A, B, C)

Sol. x = C [$\because$ After critical angle reflection starts], y = r – i = 90° – C & z = 180° – 2C

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SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **5 Q. [4 M (0)]**

18. Ans. 6

Sol. $\angle ABC = \pi - 2(\hat{i} - \hat{r})$

$2\angle ABC = \angle AOC$ [half angle theorem]

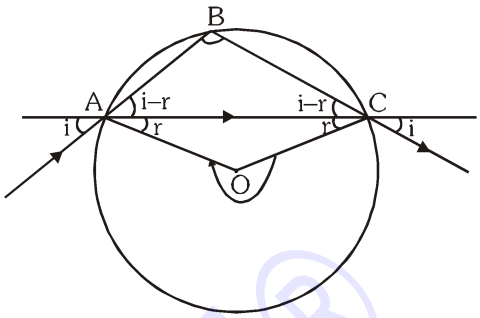
$2\pi - 4\hat{i} + 4r = (\pi + 2r)$

$2\hat{i} - r = \frac{\pi}{z} \Rightarrow r = 2\hat{i} - \frac{\pi}{2}$

$1\sin\hat{i} = \mu\sin\hat{r}$

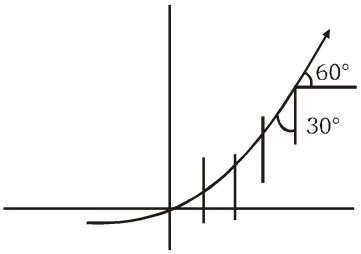
$\sin\hat{i} = \mu\sin\left(2\hat{i} - \frac{\pi}{2}\right)$

$\sin\hat{i} = \frac{\sqrt{3}}{2} \Rightarrow \hat{i} = \frac{\pi}{3} \Rightarrow \frac{2\pi}{6} \Rightarrow k = 6$



19. Ans. 1

Sol.



$\mu_1 \sin\theta_1 = c$, $1\sin 90° = (1 + y^2)\sin 30° \Rightarrow y = 1$

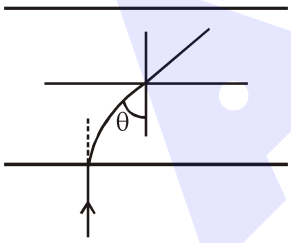
20. Ans. 5

Sol. Let frequency of rotating mirror be f.

$\therefore \frac{2D}{v} \times 2\pi f = \frac{\theta}{2} \Rightarrow D = \frac{25}{4} m$

21. Ans. 2

Sol. Apply snell's law, $\mu \sin\theta = \text{constant}$



From Snell's law

$1 = \mu\sin\theta$

$\sin\theta = \frac{1}{\mu}$

$\frac{dy}{dx} = \tan\theta = \frac{\sin\theta}{\cos\theta}$

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$$\frac{dy}{dx} = \frac{1/\mu}{\frac{\left(\mu^2-1\right)^{1/2}}{\mu}}$$

$$\frac{dy}{dx} = e^{\frac{-x}{2d}}$$

$$\int dy = \int e^{\frac{-x}{2d}}\, dx$$

$y = -2d\ e^{\frac{-x}{2d}} + C$

at $x = 0, y = 0, C = 2d$

at $y = d$

$$y = -2d\, e^{\frac{-x}{2d}} + 2d$$

$$-d = -2d\, e^{\frac{-x}{2d}}$$

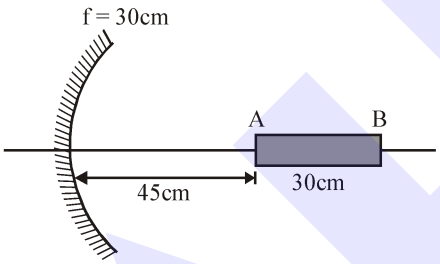
$$\frac{1}{2} = e^{\frac{-x}{2d}}$$

$$\ell n2 = \frac{x}{2d}$$

$2d\ell n2 = x$

22. Ans. 4

Sol.



Let's find the image of end A.

$$\Rightarrow \frac{1}{V_1} + \frac{1}{45} = \frac{1}{30}$$

$\Rightarrow V_1 = 90$ cm ... (i)

$\Rightarrow$ Let's find the image of end B.

$$\Rightarrow \frac{1}{V_2} + \frac{1}{(45+30)} = \frac{1}{30}$$

$V_2 = 50$ cm ... (ii)

$\Rightarrow$ So length of image $= |V_1 - V_2| = 40$ cm

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SECTION-IV

Matrix Match Type (4 × 5) **1 Q. [8 M (for each entry +2(0)]**

23. Ans. (A) – R; (B) – S; (C) – P; (D) – Q

Sol. (C) Velocity of bird in water = $6 \times \frac{4}{3} = 8 \downarrow$

w.r.t fish = 8 + 4 = 12 ↓

(D) Velocity of bird in water after reflection from mirror = 8 ↑

w.r.t fish = 8 – 4 = 4 ↑

(A) Velocity of fish in air = $4 \times \frac{3}{4} = 3 \uparrow$

Velocity of fish w.r.t bird = 3 + 6 = 9 ↑

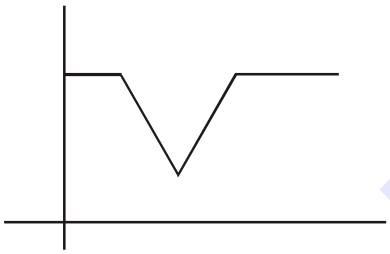
(B) Velocity of image of fish after reflection from mirror in air $= 4 \times \frac{3}{4} = 3 \downarrow$

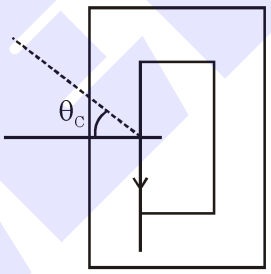
w.r.t bird = – 3 + 6 = 3 ↑

Subjective Type **4 Q. [4 M (0)]**

24. Ans. $\theta \approx$ **53°**

Sol.





For refraction from plastic sheet

$i < \theta_C$

maximum angle

$i = \theta_C$

$\sin i = \sin \theta_C = \frac{\mu_r}{\mu_D} = \frac{1.2}{1.5}$

$\sin \theta_C = \frac{4}{5}$

$\theta_C = 53°$

25. Ans. (30 cm , – 14 mm)

Sol. Reflection at concave mirror $v = \frac{uf}{u-f} = \frac{(-20)(-15)}{-20+15} = -60$ cm

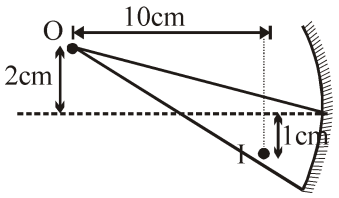
$$m = \frac{h_I}{h_0} = -\frac{v}{u} \Rightarrow h_I = -h_0 \times \left(\frac{v}{u}\right) \Rightarrow h_I = -2 \times \left(\frac{-60}{-20}\right) = -6 \text{ mm}$$

Again reflection at convex mirorr

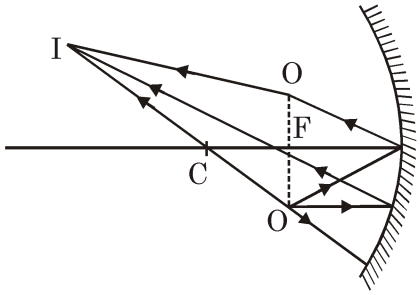
$$v = \frac{uf}{u-f} = \frac{10 \times 20}{10-20} = -20 \text{ cm} \; ; \; h_I = -h_0 \times \frac{v}{u} = -(-8) \times \left(\frac{-20}{10}\right) = -16 \text{ mm}$$

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26. **Ans.** 20 cm from object, 20/3 cm from object

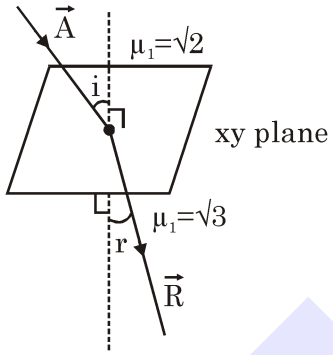


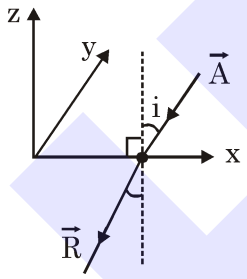
Sol.



27. **Ans.** $\vec{r}=\frac{3}{5\sqrt{2}}\hat{i}+\frac{2\sqrt{2}}{5}\hat{j}-\frac{1}{\sqrt{2}}\hat{k}\,(\text{angle of incidence}=60^0;r=45^0)$

Sol.





From the figure, it can be seen that normal to the boundary separating the two media will be in z-direction, as incident ray, refracted ray and normal are in the same plane

The angle made by the incident ray with the normal is given as

$$\cos i=\frac{A_z}{A}=\frac{10}{\sqrt{\left(6\sqrt{3}\right)^2+\left(8\sqrt{3}\right)^2+100^2}}=\frac{10}{\sqrt{108+192+100}}=\frac{10}{20}$$

$$\Rightarrow \cos i=\frac{-1}{2}\Rightarrow i=\cos^{1}\left(\frac{1}{2}\right)$$

$$\Rightarrow i=60^\circ$$

Applying Snell's law of refraction at the interface of two media

$n_1 \sin i = n_2 \sin r$

$$\sqrt{2}\sin 60^\circ=\sqrt{3}\sin r$$

$$\sin r=\frac{\sqrt{2}\times\sqrt{3}}{\sqrt{3}\times 2}=\frac{1}{\sqrt{2}}$$

$$\Rightarrow r=\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)\Rightarrow r=45^\circ$$

As the X, Y components remain unchanged, so unit vector along direction of refracted ray is

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$$\hat{R} = \frac{6\sqrt{3}\,\hat{i} + 8\sqrt{3}\,\hat{j}}{\sqrt{\left(6\sqrt{3}\right)^2 + \left(8\sqrt{3}\right)^2}} \sin 45 + \cos 45\left(-\hat{k}\right)$$

$$\hat{R} = \frac{3}{5\sqrt{2}}\hat{i} + \frac{4}{5\sqrt{2}}\hat{j} - \frac{1}{\sqrt{2}}\hat{k}$$

$$\hat{R} = \frac{1}{5\sqrt{2}}\left(3\hat{i} + 4\hat{j} - 5\hat{k}\right)$$

Hence, the answer is $\frac{1}{5\sqrt{2}}\left(3\hat{i} + 4\hat{j} - 5\hat{k}\right)$

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